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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	We Need New Laptops
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4 & SDG 8
Project / Solution Name	JobTransit
Name (Leader)	Evelyn Ang
Email & Phone (Leader)	evelynang937@gmail.com, 0167606839
University Affiliation(s)	Universiti Teknologi Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence (AI) is transforming the workforce by changing job requirements and the skills needed to remain employable. In Malaysia, 56% of professionals use AI tools at work, yet only 26% have received formal AI training, according to the *Hays Asia Salary Guide 2025*. Additionally, the Ministry of Science, Technology and Innovation has stated that more than 600,000 workers will need reskilling within three to five years.

This challenge particularly affects Computer Science students, fresh graduates, early-career developers and career switchers who must decide which technical skills to develop in a rapidly evolving job market. As AI technologies and tools advance, traditional learning pathways may become outdated, making it difficult to identify emerging opportunities, assess changing skill requirements and choose suitable career directions. Employers and educational institutions also face difficulties determining which skills should be prioritized for workforce development.

Without reliable and timely career guidance, individuals risk investing time and resources in skills that may not align with future opportunities, potentially widening skills gaps and reducing employability. Limited access to relevant training may further affect career mobility and workforce productivity.

Therefore, an adaptive, evidence-based solution is needed to connect AI and job-market trends with individual capabilities, future career opportunities and personalized reskilling pathways. By linking relevant roles and skills to the United Nations Sustainable Development Goals (SDGs), such a solution can also support lifelong learning, innovation and decent work.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

JobTransit reimagines tech career navigation and reskilling through an interactive transit metaphor ("Career Metro"), applying artificial intelligence to resolve workforce AI displacement. Within the Home onboarding terminal, candidates upload resumes in PDF, Word, or image formats. The platform applies client-side OCR and the Google Gemini API with structured JSON schemas to extract verified technical competencies, followed by an adaptive questionnaire that dynamically clarifies unmentioned skills without assuming negative ability.

The Transit module visualizes career progressions as an octilinear subway map across five Computer Science lines (Agentic AI, Data, Machine Learning, Product, and Cloud). Rather than black-box scoring, its reskilling engine generates explainable pathways featuring transparent factor weights (+/- percentage deltas), hands-on capstone projects, and HRD Corp-claimable courses.

To keep learning paths aligned with active labor demands, the Trend radar crawls real-time LinkedIn job postings to analyze hiring momentum and AI exposure indices. Meanwhile, the Catchup hub runs on serverless Firebase Functions to ingest daily tech news and uses Gemini to generate role-specific "Why It Matters" digests.

Candidates manage their verified journey through the Profile module, which issues a physics-enabled 3D Digital Transit Pass with gyroscopic mouse tilt, holographic specular reflections, and real-time Cloud Firestore synchronization.

Our architecture directly leverages tools and skills gained from the SDC Phase 1 workshops. For GenAI, we implement the Google Gemini API with schema constraints for structured resume extraction, dynamic question generation, and news synthesis. For database architecture, Google Firebase provides secure authentication via Firebase Auth and flexible NoSQL storage via Cloud Firestore. Finally, Replit serves as the cloud hosting platform for containerized deployment and team development.

3. Technical Architecture & Implementation Plan

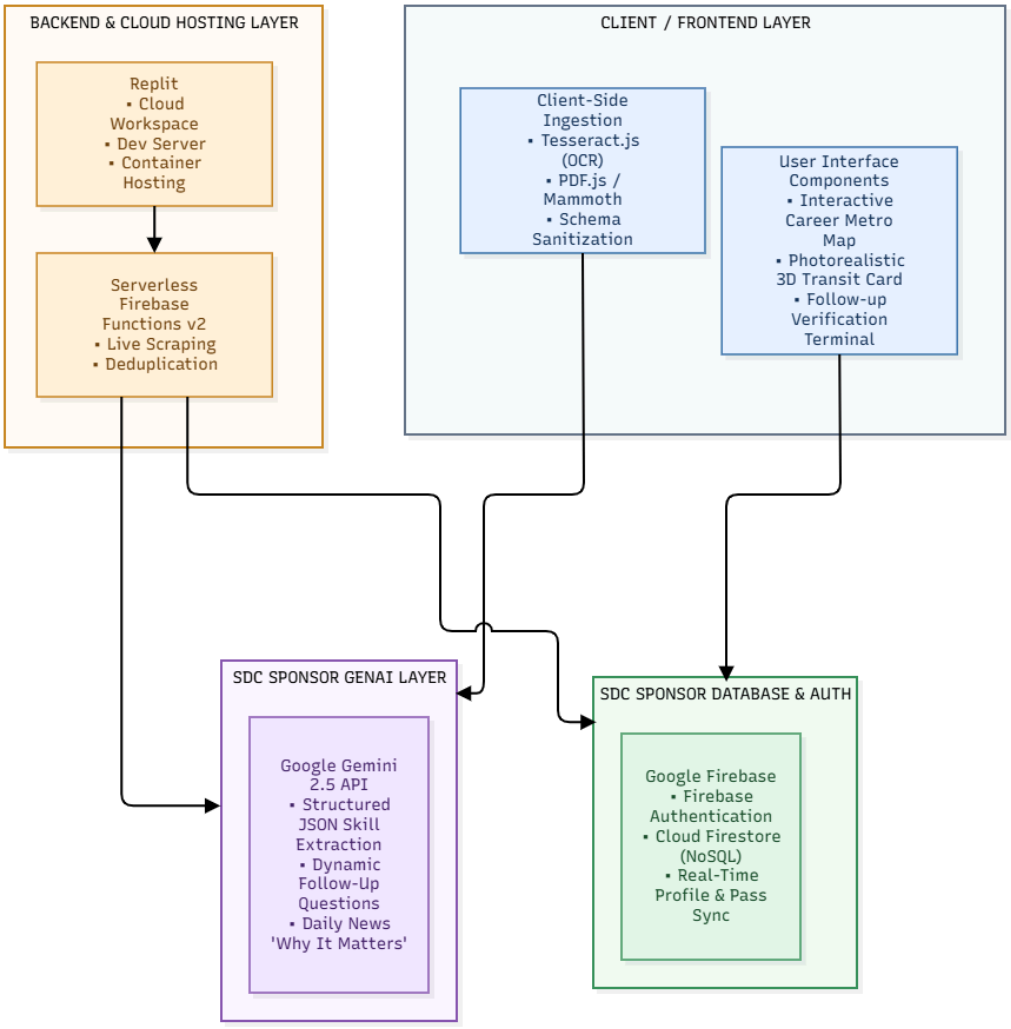
3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- Frontend / UI: React 19, TypeScript, Vite, Tailwind CSS, Lucide React
- Backend & API: Node.js / Express (or Cloud Functions) REST API executing session rules, validation logic, and profile updates.
- Database & Authentication: Google Firebase (Cloud Firestore & Firebase Auth):
    - Firebase Auth: Seamless user authentication and session management.
    - Cloud Firestore: Flexible NoSQL document database storing user profiles, question banks, answer provenances, and generated Transit Cards with real-time sync.
- GenAI Layer: GenAI API (Google Gemini API) with structured JSON schemas for CV parsing, skill extraction, and personalized question generation.
- Hosting & Deployment: Replit

System Block Diagram



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team implemented a user-centric design-thinking framework across five stages:

1. Empathize & Define
    - Surveyed early-career developers and graduates, uncovering that traditional career assessments are intimidating and penalize unmentioned skills. We defined the core principle: never assume negative proficiency.
2. Ideate
    - Conceived the "Transit" paradigm—framing reskilling not as a pass/fail exam, but as navigating train stations toward optimal destination roles.
3. Prototype
    - Developed an adaptive follow-up question machine with single-click responses and transparent provenance tags (cv vs user_confirmed), ensuring low cognitive friction.
4. Test
    - Validated multi-path candidate profiles (e.g., full-stack devs vs. non-traditional switchers) to ensure question queues dynamically prune irrelevant checkpoints.

Phase 2 Deployment Strategy:

The application uses a decoupled client-server architecture built on Vite and containerized web services. Phase 2 milestones include connecting the live GenAI parser, expanding Malaysian tech-industry job taxonomy datasets, conducting user acceptance testing (UAT) with university cohorts, and deploying to production cloud infrastructure with HTTPS and automated CI/CD pipelines.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	• Target 8.6: "By 2020, substantially reduce the proportion of youth not in employment, education or training" • Target 4.4: "Substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In Malaysia, 56% of professionals use AI tools at work, yet only 26% have received training to use them (Hays Asia Salary Guide 2025). Meanwhile, approximately 600,000 workers across 10 strategic economic sectors were identified as needing reskilling within the next 3–5 years due to AI, digitalisation and the green economy (BERNAMA, 2024). These figures highlight the need for more accessible and targeted skills-development pathways to help workers identify relevant training opportunities and prepare for changing job-market demands.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Skill Gap Clarification Latency Average time (minutes) required for a candidate to identify and confirm their critical technical skill gaps. 2. Reskilling Roadmap Relevancy Rate Percentage of generated skill milestones directly matching verified local tech hiring requirements.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Reduce the time candidates spend identifying and matching relevant upskilling courses to their career goals by 30% by the final prototype presentation, through AI-powered course recommendations, skills-gap analysis, and consolidated course information.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ✔ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ✔ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ✔ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ✔ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✔ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✔ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	**Type (code/dataset/model)**	**License**
React 19 & React-DOM	Code / Frontend Framework	MIT License
Firebase SDK	Library / Cloud Database & Auth	Apache 2.0
Vite	Code / Build Tooling	MIT License
Tailwind CSS	Code / CSS Framework	MIT License
Lucide React	Code / Icon Library	ISC License
Google GenAI SDK	Library / Model API Client	Apache 2.0
Tesseract.js	Code / OCR Library	Apache 2.0
pdfjs-dist	Code / Document Parser	Apache 2.0
mammot	Code / Document Parser	BSD-2-Clause

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✔ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



25 October 2026

8. References

AI drives decline in Software Development roles, but Malaysia holds strong as key global skill hub. (n.d.). Hays Malaysia. https://www.hays.com.my/press-release/content/future-of-work-malaysia-technology-ai-impact

Augustin, S., & Today, F. M. (2024, October 10). 600,000 employees need reskilling in 3 to 5 years, says Sim. *Free Malaysia Today | FMT*. https://www.freemalaysiatoday.com/category/nation/2024/10/10/600000-employees-need-reskilling-in-3-to-5-years-says-sim